Arpit Verma

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EDUCATION

Vellore Institute of Technology — **Sehore, M.P.**
B.Tech. in Computer Science and Engineering — *Expected May 2027 | CGPA : 8.40/10.00*

Sagar Public School — **Bhopal, India**
Class XII (77.4%) — *May 2023*

Sagar Public School — **Bhopal, India**
Class X (93.4%) — *August 2021*

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL
Backend: Spring Boot, Spring Security, Spring Data JPA, REST APIs, Node.js, Express.js
Frontend: React, TypeScript, HTML5, CSS3, Tailwind CSS, Axios
Databases: MySQL, MongoDB
AI Integration: OpenRouter API, Prompt Engineering
Tools: Git, Docker, Maven, Postman, AWS
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

Job Tracker Application | Java, Spring Boot, Spring Security, React, MySQL, JWT, Docker, REST APIs
GitHub | Live Demo

- Developed a production-ready full-stack Job Tracker using **Spring Boot, React, and MySQL** with a **layered architecture**, secure REST APIs, and authenticated multi-user access.
- Designed **REST APIs** supporting **CRUD operations, filtering, pagination, sorting, and search**, reducing response payloads while improving usability for large application datasets.
- Secured APIs using **Spring Security, JWT authentication, BCrypt hashing** and **role-based authorization**, preventing unauthorized resource access.
- **Dockerized** frontend and backend services and deployed across **Render and Vercel** to enable reproducible deployments and simplified environment configuration.

Narrative Generation System | Python, FastAPI, React, LLMs, OpenRouter API, Docker
GitHub | Live Demo

- Developed a modular AI-powered storytelling platform using **FastAPI, React (TypeScript), and OpenRouter** with provider abstraction for interchangeable LLM backends.
- Designed a **Factory/Strategy-based LLM integration layer** with an abstract provider interface, enabling seamless switching between a live OpenRouter provider and a mock provider for testing.
- Engineered **fault-tolerant API calls** with retry logic, exponential backoff, and multi-model fallback chains to gracefully handle rate limits and provider failures.
- Implemented a stateful, branching narrative engine where reader choices persist across sessions and drive RPG-style progression (custom state variables and inventory).
- Wrote **unit tests (pytest)** covering LLM provider behavior and story-specification generation to ensure core module correctness.

EXPERIENCE

Core Member – Human Resource Team
Data Science Club, VIT Bhopal — Sep 2025 – Present

- Coordinated recruitment drives involving candidate screening, scheduling, and communication across multiple teams.
- Managed **Neural Nexus**, a university-level Data Science event, collaborating across multiple teams.

ACHIEVEMENTS

- Active Core Member of the Data Science Club, VIT Bhopal.

CERTIFICATIONS

- **NPTEL – Introduction to Machine Learning**
- **NPTEL – Marketing Analytics (Elite + Topper, Top 1**
- **ServiceNow Virtual Internship Program - SmartBridge**
- **Coursera – The Bits and Bytes of Computer Networking**